

Heuristic Search Course Project Instructions

The course project is a mandatory part of the course and counts for **30% of the final course grade**. The final exam counts for the remaining **70%**.

The goal of the project is to study a published research paper on heuristic search, reproduce its central experimental results, and meaningfully extend the work.

1. Project Topic

Each team should choose a **published paper** from a recognized conference or journal related to heuristic search. Relevant venues include AAAI, IJCAI, ICAPS, AAMAS, SoCS, AIJ, JAIR, JAAMAS, and other relevant conferences and journals.

The selected paper must be approved before the team starts working on the project.

Possible topics include, but are not limited to:

- heuristic search algorithms
- heuristic functions
- bounded-suboptimal search
- anytime search
- bidirectional heuristic search
- multi-objective search
- multi-agent path finding
- pathfinding and graph search
- search-based optimization
- other closely related topics, subject to approval

2. Project Approval

Each project must be approved by email.

For approval, each team should send an email containing:

1. the title of the selected paper
2. the full bibliographic details of the paper
3. the team members
4. a short description of the planned reproduction
5. a short description of the planned extension/contribution

Projects that are not approved may not be submitted as course projects

3. Team Size and Contribution

Projects should be performed in teams of **two or three students**, unless approved otherwise.

Students in the same team are expected to contribute substantially to the project. The report must state how each student contributed to the project.

In the project defense, students from the same team may receive different grades, depending on their demonstrated understanding and contribution.

4. Project Requirements

Each project must include the following components.

4.1 Reproduction of Results

The team must reproduce at least **two central experimental results** from the selected paper.

The reproduced results may be based on code provided by the authors of the original paper, or on an independent implementation by the team.

The report must clearly explain:

- which results from the original paper were reproduced
- how the experiments were reproduced
- whether the reproduced results match the original results
- any discrepancies between the reproduced and original results
- possible reasons for these discrepancies

Students are not required to reproduce every experiment in the original paper. However, the reproduced results must be central enough to demonstrate a meaningful understanding and evaluation of the selected paper.

4.2 Project Extension / Contribution

In addition to reproducing results from the selected paper, each team must add a new contribution or extension.

Acceptable extensions include, for example:

- comparison to another algorithm
- evaluation on a new benchmark domain
- development of a new heuristic
- development of a new method, policy, or variant
- evaluation of a different tie-breaking policy
- ablation study
- scalability analysis

- implementation improvement
- additional experimental analysis
- another relevant extension, subject to approval

The project grade will be affected by the extent, quality, and difficulty of the project contribution.

4.3 Scientific Report

Each team must submit a scientific report describing the project.

The report must be:

- written solely in **English**
- at most **6 pages**, not including references
- written in **Times New Roman**, font size **12**
- submitted as a PDF file.

Additional results may be submitted together with the report as an appendix. The appendix is not counted as part of the 6-page limit.

The report should be written in a scientific style. Possible sections include:

1. Abstract
2. Introduction
3. Selected Paper Description
4. Project Description
5. Experiments
6. Discussion
7. Conclusion
8. References

The exact section structure may be changed if appropriate for the project, but the report must clearly describe the selected paper, the reproduced results, the new contribution, the experimental setup, the results, and the conclusions.

The report must include full details of the experiments performed, including:

- hardware used
- software used
- programming language and relevant libraries
- benchmark instances
- algorithms and configurations

- parameter settings
- time and memory limits
- number of runs, if relevant
- random seeds, if relevant
- any differences from the experimental setup of the original paper

4.4 Reproducibility Statement

The report must include a clear reproducibility statement.

The reproducibility statement should explicitly state:

- which results from the original paper were reproduced
- which results from the original paper were not reproduced
- whether the implementation was based on provided code or on a new implementation
- what changes were made compared to the original paper
- whether the reproduced results agree with the original results
- what files are included in the submitted implementation and raw results
- how the submitted code and experiments can be run

4.5 Use of AI Tools

Using AI tools is allowed.

However, the report must explicitly state which AI tools were used during the project and how. This includes, for example, using AI tools for writing assistance, code generation, debugging, explanations, or data analysis.

The team is fully responsible for the correctness, originality, and scientific quality of the submitted work.

4.6 Recorded Video

Each team must submit a recorded video presenting the project.

The video must be at most **8 minutes long**. This limit is strict.

The video should clearly present:

- the selected paper
- the motivation for the project
- the reproduced results
- the new contribution or extension
- the experimental setup

- the main results
- the conclusions

Not all team members are required to speak in the video. However, the speaker must be a team member, and the video may not use a generated voice instead of a team member speaking.

It is recommended to record the video using Zoom, but this is not mandatory.

The text appearing in the video must be in **English**. The speaker may speak in either **English or Hebrew**.

4.7 Project Defense

At the end of the semester, each team will defend its project and answer questions.

The defenses will take place on:

- **June 25**
- **June 28**

Selecting specific time slots for the defenses will be possible as we get closer to the defense dates. The time slots will be picked by the students.

Students for whom these dates are not possible should schedule a different defense time by email.

More details about the defense format will be published closer to the defense dates. During the defense, students may be asked questions individually. Therefore, each student must be familiar with all central parts of the project, including the selected paper, the implementation, the experiments, the results, and the project contribution.

5. Submission

The project must be submitted as a single ZIP file.

The ZIP file must contain:

1. the scientific report as a PDF file
2. the recorded video
3. the implementation
4. instructions for running the code
5. configuration files, if relevant
6. the raw results files
7. any additional files needed to understand or reproduce the experiments

The submitted files should be organized clearly.

6. Deadline

The deadline for submitting the project is: **June 23 at 23:55** (unless approved otherwise).

7. Grading

The project counts for **30% of the final course grade**.

The project grade will be divided as follows:

- Report: **30%**
- Reproduction of results: **15%**
- New contribution/extension: **15%**
- Recorded video: **10%**
- Defense: **30%**

In the defense, students from the same team may receive different grades.

The grade will reflect both the submitted material and the students' demonstrated understanding of the selected paper, the reproduced results, the implementation, the experiments, and the new contribution.